

## Import Libraries

```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

import statsmodels.api as sm
from statsmodels.stats.outliers_influence import variance_inflation_factor
from statsmodels.stats.diagnostic import het_breuschpagan
from scipy.stats import shapiro

sns.set_style("whitegrid")

plt.rcParams["figure.figsize"] = (8,5)
```

## Load Dataset

```
df = pd.read_csv("/content/drive/MyDrive/Datasets/marketing_sales_data.csv")

print(df.head())
```

|   | TV     | Radio     | Social Media | Influencer | Sales      |
|---|--------|-----------|--------------|------------|------------|
| 0 | Low    | 3.518070  | 2.293790     | Micro      | 55.261284  |
| 1 | Low    | 7.756876  | 2.572287     | Mega       | 67.574904  |
| 2 | High   | 20.348988 | 1.227180     | Micro      | 272.250108 |
| 3 | Medium | 20.108487 | 2.728374     | Mega       | 195.102176 |
| 4 | High   | 31.653200 | 7.776978     | Nano       | 273.960377 |

## Dataset Information

```
print(df.info())

print(df.shape)

print(df.describe(include="all"))
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 572 entries, 0 to 571
Data columns (total 5 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   TV              572 non-null    object
1   Radio           572 non-null    float64
2   Social Media    572 non-null    float64
3   Influencer      572 non-null    object
4   Sales           572 non-null    float64
dtypes: float64(3), object(2)
memory usage: 22.5+ KB
None
(572, 5)
```

|        | TV     | Radio      | Social Media | Influencer | Sales      |
|--------|--------|------------|--------------|------------|------------|
| count  | 572    | 572.000000 | 572.000000   | 572        | 572.000000 |
| unique | 3      | NaN        | NaN          | 4          | NaN        |
| top    | Medium | NaN        | NaN          | Nano       | NaN        |
| freq   | 205    | NaN        | NaN          | 157        | NaN        |
| mean   | NaN    | 17.520616  | 3.333803     | NaN        | 189.296908 |
| std    | NaN    | 9.290933   | 2.238378     | NaN        | 89.871581  |
| min    | NaN    | 0.109106   | 0.000031     | NaN        | 33.509810  |
| 25%    | NaN    | 10.699556  | 1.585549     | NaN        | 118.718722 |
| 50%    | NaN    | 17.149517  | 3.150111     | NaN        | 184.005362 |
| 75%    | NaN    | 24.606396  | 4.730408     | NaN        | 264.500118 |
| max    | NaN    | 42.271579  | 11.403625    | NaN        | 357.788195 |

## Missing Values

```
print(df.isnull().sum())
```

```
TV          0
Radio       0
Social Media 0
Influencer  0
Sales       0
dtype: int64
```

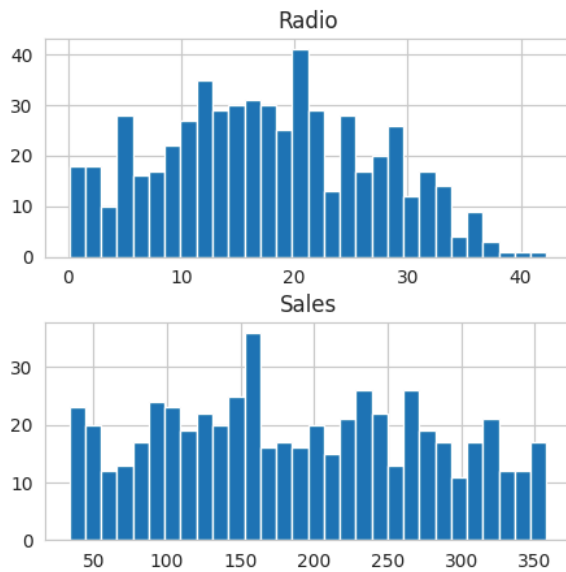
## Exploratory Data Analysis:

## Numerical Variables

```
numeric = df.select_dtypes(include=np.number)

numeric.hist(figsize=(12,5), bins=30)

plt.show()
```

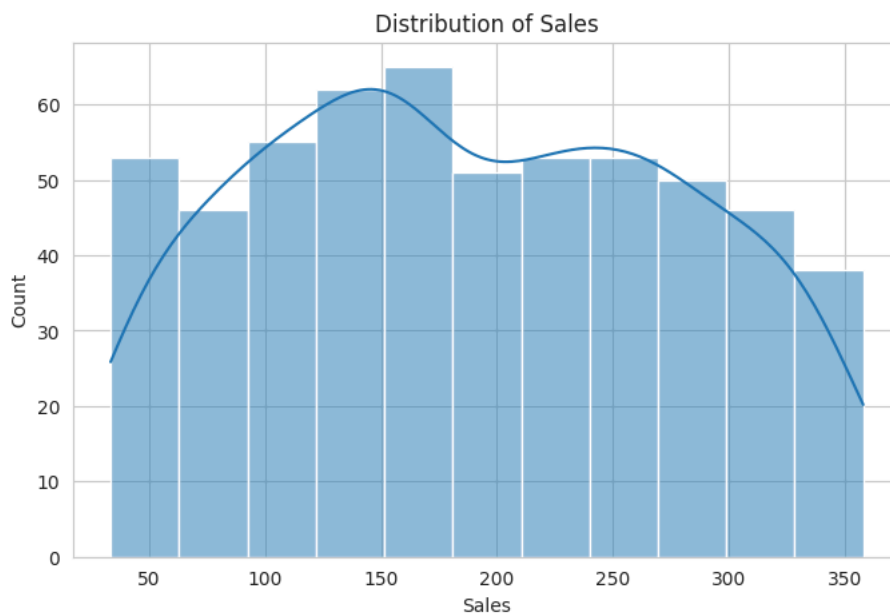


## Distribution of Sales

```
sns.histplot(df["Sales"], kde=True)

plt.title("Distribution of Sales")

plt.show()
```

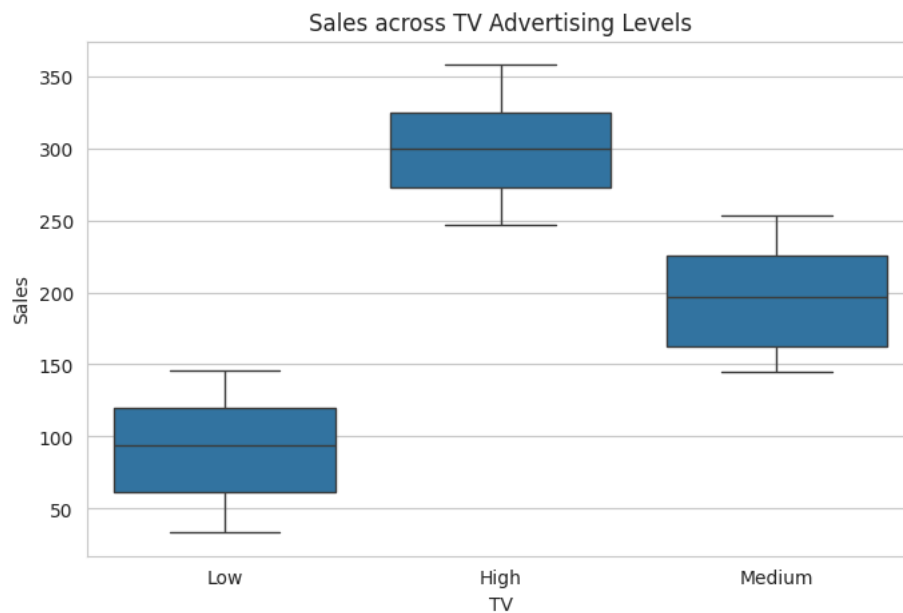


## TV vs Sales

```
sns.boxplot(
    data=df,
    x="TV",
    y="Sales"
)

plt.title("Sales across TV Advertising Levels")

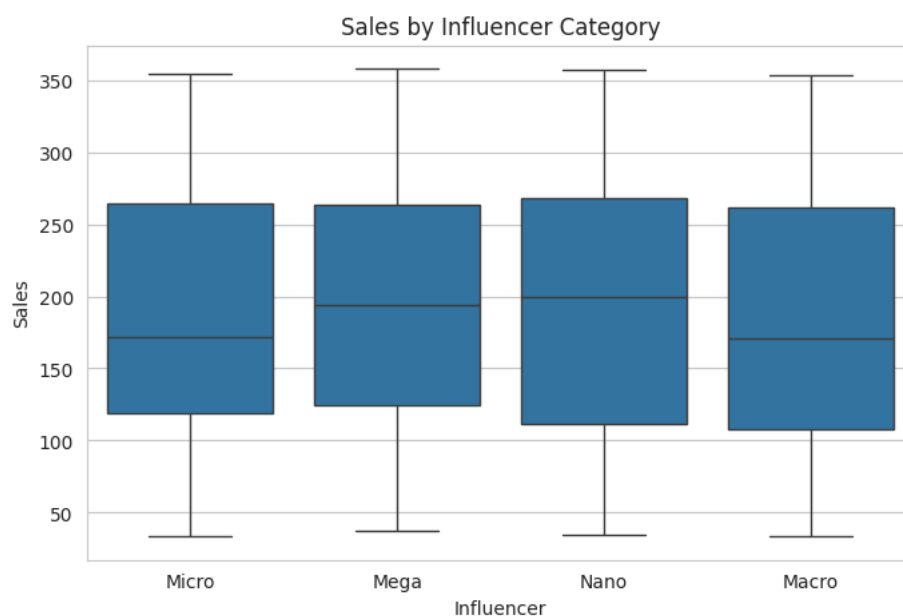
plt.show()
```



Influencer vs Sales

```
sns.boxplot(
    data=df,
    x="Influencer",
    y="Sales"
)

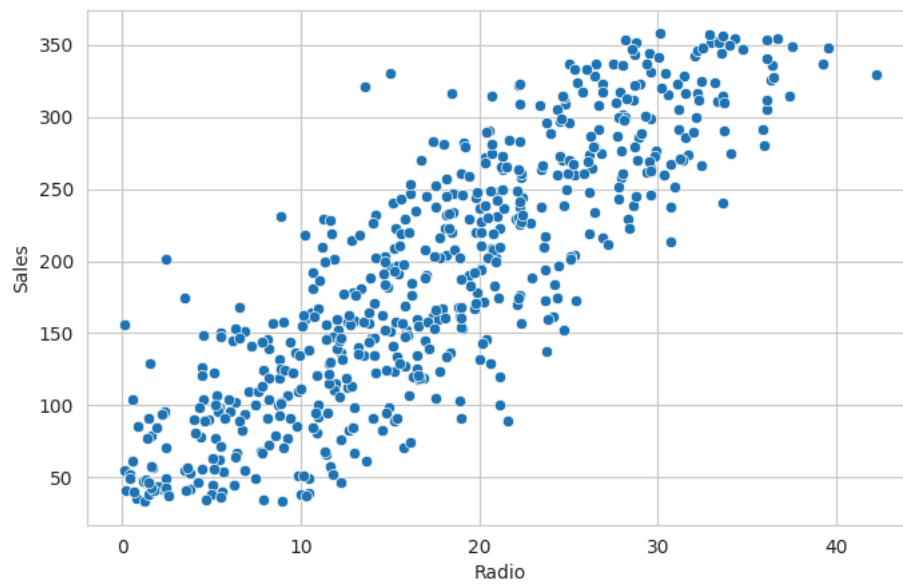
plt.title("Sales by Influencer Category")
plt.show()
```



Radio

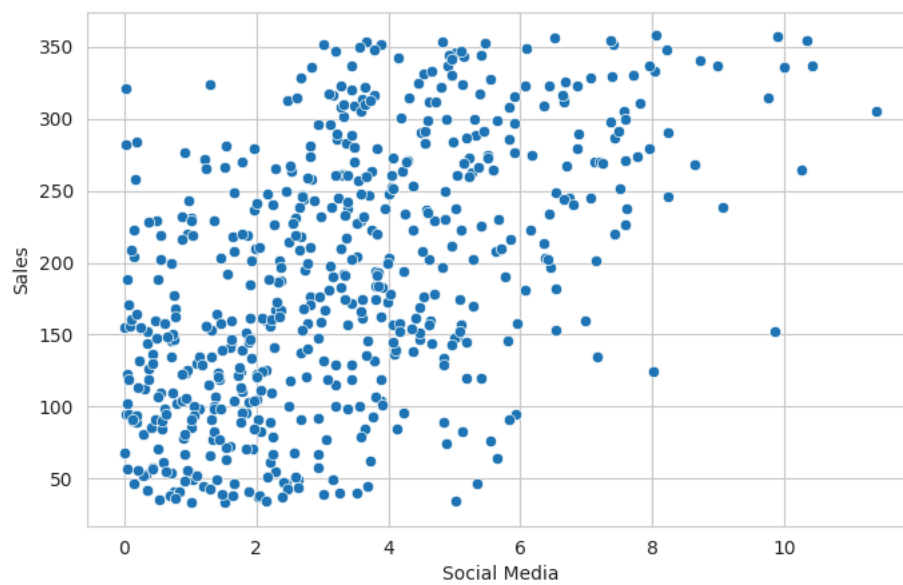
```
sns.scatterplot(
    data=df,
    x="Radio",
    y="Sales"
)

plt.show()
```



Social Media

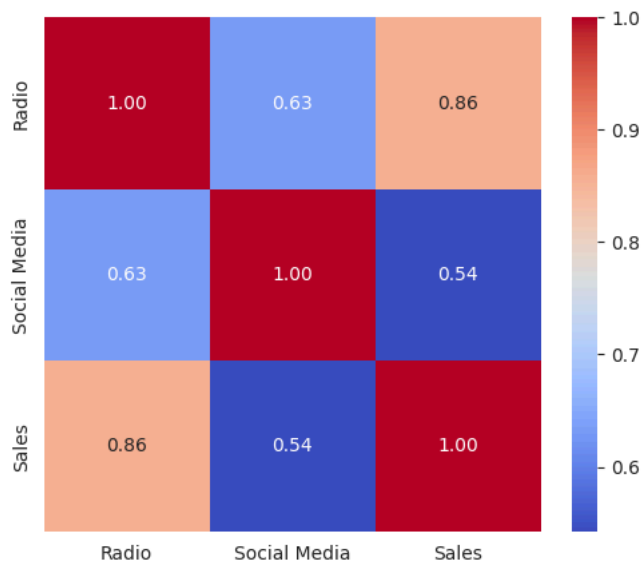
```
sns.scatterplot(  
    data=df,  
    x="Social Media",  
    y="Sales"  
)  
  
plt.show()
```



Correlation Matrix

Only numerical variables are included.

```
plt.figure(figsize=(6,5))  
  
sns.heatmap(  
    numeric.corr(),  
    annot=True,  
    cmap="coolwarm",  
    fmt=".2f"  
)  
  
plt.show()
```



Notice that Radio has a stronger relationship with Sales than Social Media.

Convert Categorical Variables

Since TV and Influencer are categorical, create dummy variables.

```
df_encoded = pd.get_dummies(
    df,
    columns=["TV", "Influencer"],
    drop_first=True
)
```

View the transformed dataset.

```
print(df_encoded.head())
```

|   | Radio     | Social Media | Sales      | TV_Low | TV_Medium | Influencer_Mega \ |
|---|-----------|--------------|------------|--------|-----------|-------------------|
| 0 | 3.518070  | 2.293790     | 55.261284  | True   | False     | False             |
| 1 | 7.756876  | 2.572287     | 67.574904  | True   | False     | True              |
| 2 | 20.348988 | 1.227180     | 272.250108 | False  | False     | False             |
| 3 | 20.108487 | 2.728374     | 195.102176 | False  | True      | True              |
| 4 | 31.653200 | 7.776978     | 273.960377 | False  | False     | False             |

|   | Influencer_Micro | Influencer_Nano |
|---|------------------|-----------------|
| 0 | True             | False           |
| 1 | False            | False           |
| 2 | True             | False           |
| 3 | False            | False           |
| 4 | False            | True            |

Define Predictor Matrix

```
X = df_encoded.drop("Sales", axis=1)
y = df_encoded["Sales"]
```

Add Constant

```
X = sm.add_constant(X)
```

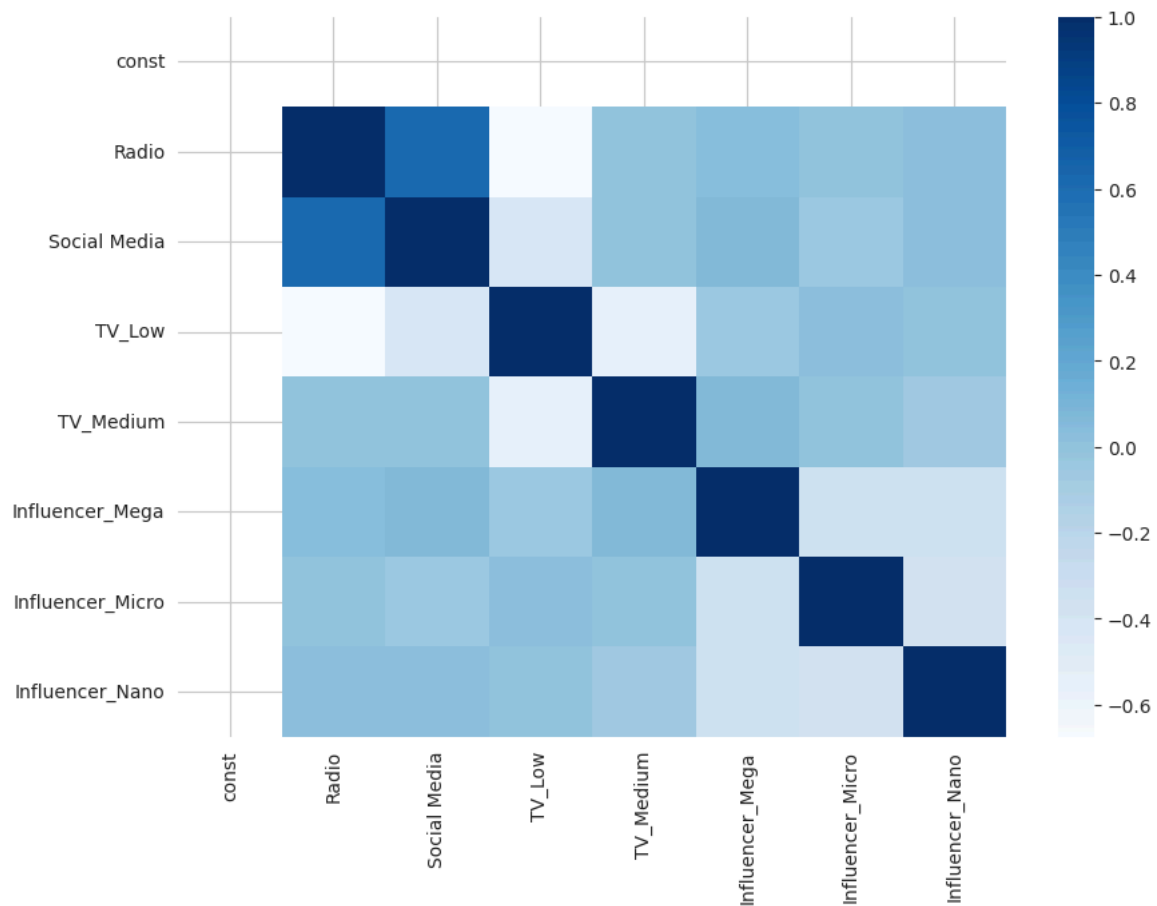
Multicollinearity

Correlation Matrix

```
plt.figure(figsize=(10,7))

sns.heatmap(
    X.corr(),
    cmap="Blues"
)

plt.show()
```



```
print(X.dtypes)
```

```
const          float64
Radio          float64
Social Media   float64
TV_Low         bool
TV_Medium      bool
Influencer_Mega bool
Influencer_Micro bool
Influencer_Nano bool
dtype: object
```

Convert everything to numeric

```
X = X.astype(float)
```

compute the VIF

Variance Inflation Factor

```
vif = pd.DataFrame()

vif["Variable"] = X.columns

vif["VIF"] = [
    variance_inflation_factor(X.values, i)
    for i in range(X.shape[1])
]

print(vif)
```

```
Variable  VIF
0      const 31.649565
1      Radio  3.479641
2  Social Media 1.669098
3      TV_Low  4.076384
4    TV_Medium  2.233350
5  Influencer_Mega 1.593889
6  Influencer_Micro 1.618434
7  Influencer_Nano 1.627430
```

## Multiple Linear Regression

```
model = sm.OLS(y, X).fit()

print(model.summary())
```

```
=====
                        OLS Regression Results
=====
Dep. Variable:          Sales      R-squared:                0.904
Model:                  OLS       Adj. R-squared:           0.903
Method:                 Least Squares   F-statistic:             760.4
Date:                  Sun, 28 Jun 2026   Prob (F-statistic):      1.82e-282
Time:                  22:25:10      Log-Likelihood:          -2713.4
No. Observations:      572          AIC:                    5443.
Df Residuals:          564          BIC:                    5478.
Df Model:              7
Covariance Type:       nonrobust
=====
                        coef      std err          t      P>|t|      [0.025      0.975]
-----
const                217.4784      6.584      33.031      0.000      204.546      230.411
Radio                 2.9735      0.235      12.644      0.000       2.512       3.435
Social Media        -0.1391      0.676      -0.206      0.837      -1.467       1.189
TV_Low             -154.5736      4.949     -31.231      0.000     -164.295     -144.852
TV_Medium          -75.5947      3.647     -20.726      0.000     -82.759     -68.431
Influencer_Mega      2.4948      3.462       0.721      0.471      -4.305       9.295
Influencer_Micro     2.9391      3.378       0.870      0.385      -3.695       9.574
Influencer_Nano       0.8015      3.346       0.240      0.811      -5.770       7.373
=====
Omnibus:              58.711   Durbin-Watson:           1.874
Prob(Omnibus):        0.000   Jarque-Bera (JB):        17.802
Skew:                 0.057   Prob(JB):                0.000136
Kurtosis:             2.143   Cond. No.:               147.
=====

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```

## Model Performance

```
print("R-squared:", model.rsquared)

print("Adjusted R-squared:", model.rsquared_adj)

print(model.pvalues)

print(model.params)
```

```
R-squared: 0.9041902290184094
Adjusted R-squared: 0.9030011006551627
const                5.906914e-134
Radio                1.943682e-32
Social Media         8.371166e-01
TV_Low              4.536854e-125
TV_Medium           2.277532e-71
Influencer_Mega      4.714413e-01
Influencer_Micro     3.845913e-01
Influencer_Nano      8.107454e-01
dtype: float64
const                217.478405
Radio                2.973532
Social Media         -0.139054
TV_Low              -154.573626
TV_Medium           -75.594662
Influencer_Mega      2.494795
Influencer_Micro     2.939094
Influencer_Nano       0.801541
dtype: float64
```

## Predictions

```
predictions = model.predict(X)

df_encoded["Predicted Sales"] = predictions
```

## Residuals

```
residuals = model.resid
```

## Diagnostic Plot

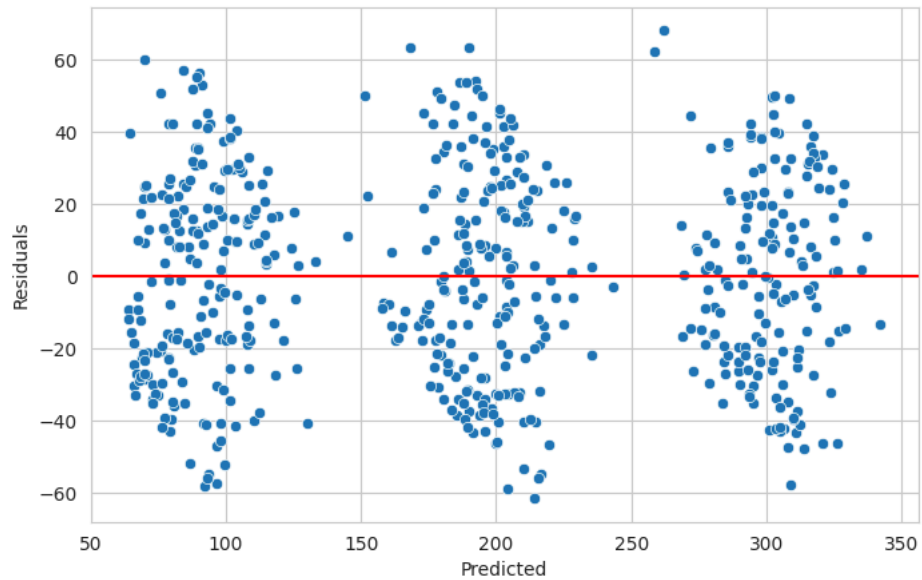
```
sns.scatterplot(
    x=predictions,
    y=residuals
)

plt.axhline(0,color="red")

plt.xlabel("Predicted")

plt.ylabel("Residuals")

plt.show()
```

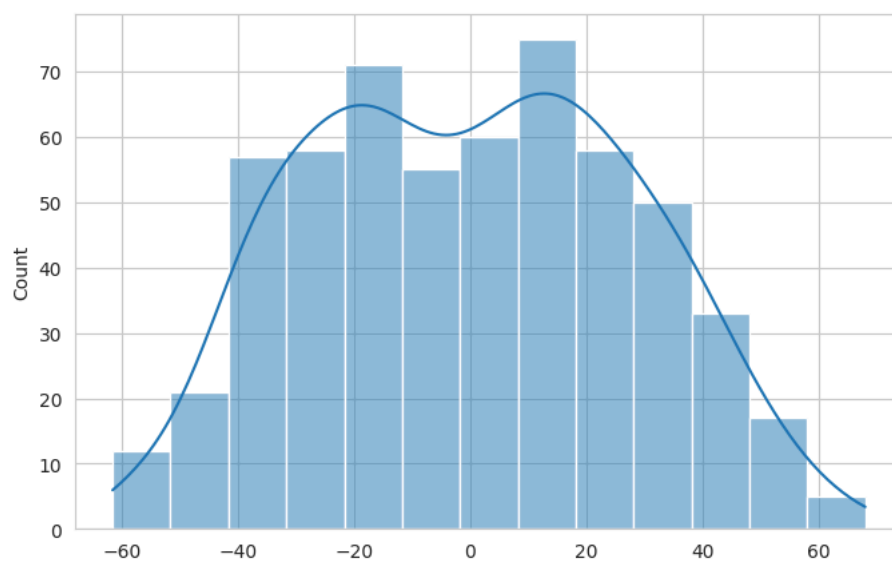


Normality

Histogram

```
sns.histplot(
    residuals,
    kde=True
)

plt.show()
```

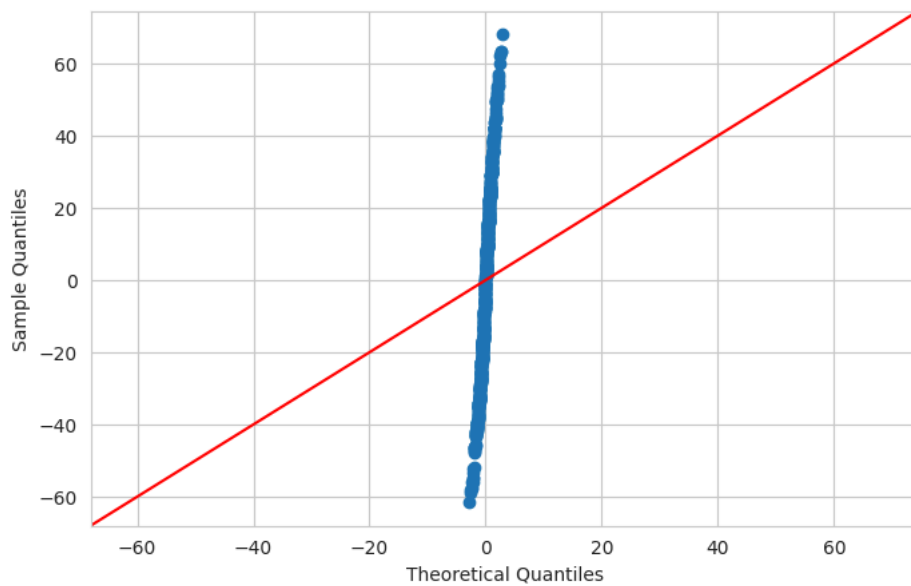


QQ Plot

```
sm.qqplot(
    residuals,
    line="45"
)
```



```
plt.show()
```



Shapiro-Wilk

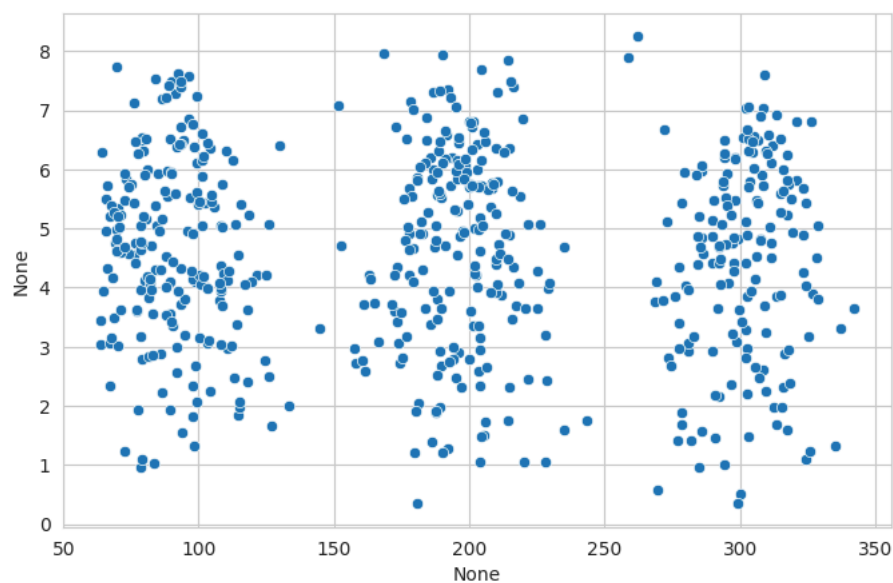
```
stat,p = shapiro(residuals)
print("Statistic:",stat)
print("P-value:",p)
```

```
Statistic: 0.9834225140471117
P-value: 4.301684691826888e-06
```

Homoscedasticity

Scale Location Plot

```
sns.scatterplot(
    x=predictions,
    y=np.sqrt(np.abs(residuals))
)
plt.show()
```



Breusch-Pagan

```
bp = het_breuschpagan(
    residuals,
```

```

X
)

labels = [
    "LM Statistic",
    "LM p-value",
    "F Statistic",
    "F p-value"
]

print(dict(zip(labels,bp)))

{'LM Statistic': np.float64(8.532947953442553), 'LM p-value': np.float64(0.2879487100484683), 'F Statistic': np.float64(1.22

```

## Actual vs Predicted

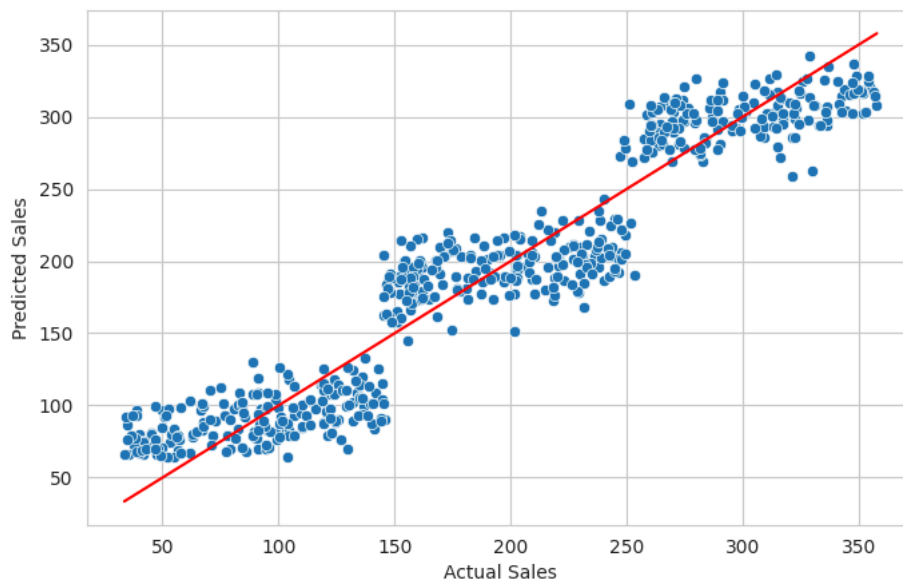
```

sns.scatterplot(
    x=y,
    y=predictions
)

plt.plot(
    [y.min(),y.max()],
    [y.min(),y.max()],
    color="red"
)

plt.xlabel("Actual Sales")
plt.ylabel("Predicted Sales")
plt.show()

```



## Interpretation

The Adjusted  $R^2$  indicates the proportion of variability in Sales explained collectively by TV advertising level, Radio spending, Social Media spending, and Influencer category. A high Adjusted  $R^2$  suggests that the model provides a strong fit while accounting for the number of predictors included.

## Statistical Significance

The p-values associated with each predictor indicate whether that variable contributes significantly to predicting Sales after controlling for the effects of all other variables. Predictors with p-values below 0.05 are considered statistically significant.

## Coefficient Interpretation

For Radio and Social Media (continuous variables):

Holding all other variables constant, a one-unit increase in Radio advertising expenditure is associated with an average change in Sales equal to the estimated Radio coefficient. Holding all other variables constant, a one-unit increase in Social Media expenditure is associated with an average change in Sales equal to the estimated Social Media coefficient.

For TV and Influencer (categorical variables represented by dummy variables):

Each TV coefficient represents the expected difference in Sales between that advertising level and the reference category (e.g., High, if drop\_first=True was used), holding all other variables constant. Each Influencer coefficient represents the expected difference in Sales between that influencer category and the reference influencer category, holding all other variables constant.

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>Business Recommendation</p> <p>The multiple linear regression analysis demonstrates that market performance is influenced by both advertising expenditure and campaign characteristics. Radio advertising shows a strong positive relationship with Sales, while Social Media contributes to a lesser extent. The categorical variables indicate that different TV advertising levels and influencer categories have distinct impacts on Sales compared with their respective reference categories.</p> <p>Marketing managers should prioritize investment in statistically significant channels identified by the model. Budget should be directed toward the TV advertising level and influencer category associated with the greatest positive effect on Sales, while maintaining or increasing spending on Radio if its coefficient remains positive and statistically significant. Social Media investment should be evaluated based on its statistical significance and practical effect size. This evidence-based allocation strategy enables the company to maximize expected sales while using marketing resources efficiently.</p> | <p>Business Recommendation</p> <p>The multiple linear regression analysis demonstrates that marketing performance is influenced by both advertising expenditure and campaign characteristics. Radio advertising shows a strong positive relationship with Sales, while Social Media contributes to a lesser extent. The categorical variables indicate that different TV advertising levels and influencer categories have distinct impacts on Sales compared with their respective reference categories.</p> <p>Marketing managers should prioritize investment in statistically significant channels identified by the model. Budget should be directed toward the TV advertising level and influencer category associated with the greatest positive effect on Sales, while maintaining or increasing spending on Radio if its coefficient remains positive and statistically significant. Social Media investment should be evaluated based on its statistical significance and practical effect size. This evidence-based allocation strategy enables the company to maximize expected sales while using marketing resources efficiently.</p> |
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